

5.1-12.* Reconsider the model in Prob. 4.6-9. Now you are given the information that the basic variables in the optimal solution are x_2 and x_3 . Use this information to identify a system of three constraint boundary equations whose simultaneous solution must be this optimal solution. Then solve this system of equations to obtain this solution.

Minimize $Z = 3x_1 + 2x_2 + 4x_3$,

subject to

$$2x_1 + x_2 + 3x_3 = 60$$

$$3x_1 + 3x_2 + 5x_3 \geq 120$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

$$\text{Min } Z = 3x_1 + 2x_2 + 4x_3 + M\bar{x}_4 + M\bar{x}_5$$

$$\textcircled{1} \quad 2x_1 + x_2 + 3x_3 + \bar{x}_4 = 60$$

$$\textcircled{2} \quad 3x_1 + 3x_2 + 5x_3 - s_1 + \bar{x}_5 = 120$$

$$\textcircled{3} \quad x_1 = 0$$

$$Z - 3x_1 - 2x_2 - 4x_3 - M\bar{x}_4 - M\bar{x}_5 = 0$$

$$Z + (5M-3)x_1 + (4M-2)x_2 + (8M-4)x_3 - Ms_1 = 180M$$

	Z	x_1	x_2	x_3	s_1	$\bar{x}_4$	$\bar{x}_5$	RHS
Z	1	$5M-3$	$4M-2$	$8M-4$	$-M$	0	0	$180M$
$\bar{x}_4$	0	2	1	3	0	1	0	60 $\frac{60}{1} = 60$
$\bar{x}_5$	0	3	3	5	-1	0	1	120 $\frac{120}{1} = 120$

$\bar{x}_4/3$ enter x_3 leave $\bar{x}_4$

↓

	Z	x_1	x_2	x_3	s_1	$\bar{x}_4$	$\bar{x}_5$	RHS
Z	1	$\frac{-M-1}{3}$	$\frac{4M-2}{3}$	0	$-M$	$\frac{-8M+4}{3}$	0	$20M+80$
x_3	0	$2/3$	$1/3$	1	0	$1/3$	0	20 $20 \cdot 3 = 60$
$\bar{x}_5$	0	$-1/3$	$4/3$	0	-1	$-5/3$	1	20 $20 \cdot 3 = 60$

enter x_2 leave $\bar{x}_5$

	Z	x_1	x_2	x_3	s_1	$\bar{x}_4$	$\bar{x}_5$	RHS
Z	1	$-1/2$	0	0	$-1/2$	$-M+1/2$	$-M+1/2$	90
x_3	0	$3/4$	0	1	$1/4$	$3/4$	$-1/4$	15
x_2	0	$-1/4$	1	0	$-3/4$	$-5/4$	$3/4$	15

optimal: $x_1 = 0, \quad x_2 = 15, \quad x_3 = 15$
 $\text{Min } Z = 90$

5.1-13. Reconsider Prob. 4.3-6. Now use the given information and the theory of the simplex method to identify a system of three constraint boundary equations (in x_1, x_2, x_3) whose simultaneous solution must be the optimal solution, without applying the simplex method. Solve this system of equations to

$$\text{Maximize } Z = 5x_1 + 3x_2 + 4x_3,$$

subject to

$$2x_1 + x_2 + x_3 \leq 20$$

$$3x_1 + x_2 + 2x_3 \leq 30$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

find the optimal solution.

$$2x_1 + x_2 + x_3 = 20$$

$$3x_1 + x_2 + 2x_3 = 30$$

test $x_1 = 0$

$$x_2 + x_3 = 20$$

$$x_2 + 2x_3 = 30$$

$$x_1 = 0, x_2 = 10, x_3 = 10$$

$$Z = 70$$

test $x_2 = 0$

$$2x_1 + x_3 = 20$$

$$3x_1 + 2x_3 = 30$$

$$x_1 = 10$$

$$x_3 = 10 \quad x_2 = 0$$

not optimal

$x_3 = 0$ also not optimal

5.3-7. Most of the description of the fundamental insight presented in Sec. 5.3 assumes that the problem is in our standard form. Now consider each of the following other forms, where the additional adjustments in the initialization step are those presented in Sec. 4.6, including the use of artificial variables and the Big M method where appropriate. Describe the resulting adjustments in the fundamental insight.

(a) Equality constraints

(b) Functional constraints in $_$ form

(c) Negative right-hand sides

(d) Variables allowed to be negative (with no lower bound)

a.) If a constraint is:

$$a_i x = b_i$$

there is no slack variable available so we introduce an artificial variable:

$$a_i x + A_i = b_i$$

b.) If a constraint is: $a_i x \geq b_i$

we convert to $a_i x - s_i = b_i$

where $s_i \geq 0$ is a slack variable

but we also need to add an artificial variable to give a starting basic feasible solution:

$$a_i x - s_i + A_i = b_i$$

c.) If $a_i x \leq b_i$ and $b_i < 0$

we can multiply the entire constraint by -1 to make the RHS positive

$$a_i x \leq -b_i \Rightarrow -a_i x \geq b_i$$

d.) An unrestricted variable x_j is replaced by the difference of two non-negative variables

$$x_j = x_j^+ - x_j^-$$

$$x_j^+, x_j^- \geq 0$$

5.3-8. Reconsider the model in Prob. 4.6-5. Use artificial variables and the Big M method to construct the complete first simplex tableau for the simplex method, and then identify the columns that will contain S^* for applying the fundamental insight in the final tableau. Explain why these are the appropriate columns.

Maximize $Z = 5x_1 + 4x_2,$

subject to

$$3x_1 + 2x_2 \leq 6$$

$$2x_1 - x_2 \geq 6$$

and

$$x_1 \geq 0, \quad x_2 \geq 0.$$

$$3x_1 + 2x_2 + x_3 = 6$$

$$2x_1 - x_2 - x_4 + \bar{x}_5 = 6$$

$$\text{Max } Z = 5x_1 + 4x_2 - M\bar{x}_5$$

$$Z - 5x_1 - 4x_2 + M\bar{x}_5 = 0$$

$$Z + (-2M-5)x_1 + (M-4)x_2 + M = -6M$$

	Z	x_1	x_2	x_3	x_4	$\bar{x}_5$	RHS
Z	1	$-2M-5$	$M-4$	0	M	0	$-6M$
x_3	0	3	2	1	0	0	6
$\bar{x}_5$	0	2	-1	0	-1	1	6

S^* is x_3 and $\bar{x}_5$ column

$$S^* \text{ matrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

S^* matrix tracks the cumulative row operations.
 S^* must start as identity matrix.

D 5.3-1.* Consider the following problem.

$$\text{Maximize } Z = x_1 - x_2 + 2x_3,$$

subject to

$$2x_1 - 2x_2 + 3x_3 \leq 5$$

$$x_1 + x_2 - x_3 \leq 3$$

$$x_1 - x_2 + x_3 \leq 2$$

and

$$x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.$$

Let $x_4, x_5,$ and x_6 denote the slack variables for the respective constraints. After you apply the simplex method, a portion of the final simplex tableau is as follows:

Basic Variable	Eq.	Coefficient of:							Right Side
		Z	x_1	x_2	x_3	x_4	x_5	x_6	
Z	(0)	1				1	1	0	
x_2	(1)	0				1	3	0	
x_6	(2)	0				0	1	1	
x_3	(3)	0				1	2	0	

- (a) Use the fundamental insight presented in Sec. 5.3 to identify the missing numbers in the final simplex tableau. Show your calculations.
- (b) Identify the defining equations of the CPF solution corresponding to the optimal BF solution in the final simplex tableau.

$$2x_1 - 2x_2 + 3x_3 + x_4 = 5$$

$$x_1 + x_2 - x_3 + x_5 = 3$$

$$x_1 - x_2 + x_3 + x_6 = 2$$

$$c = [1 \quad -1 \quad 2]$$

$$b = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$s^* = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix}$$

$$y^* = [1 \quad 1 \quad 0]$$

$$b^* = s^* \cdot b = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 14 \\ 5 \\ 11 \end{bmatrix}$$

$$A_1 = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$z_{x_2} = y \cdot A_2 - c_2 = 0 \quad z^* = y^* \cdot b = [1 \quad 1 \quad 0] \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} = 8$$

$$A_2^* = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 14 \\ 5 \\ 11 \end{bmatrix}$$

$$A_1^* = s^* \cdot A_1 = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 4 \end{bmatrix}$$

$$A_3^* = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$z_{x_3} = y^* \cdot A_3 - c_3 = 0$$

final z_{x_1}

$$b^* \cdot A_1 - c_1 = [1 \quad 1 \quad 0] \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} = 2$$

Basic Variable	Eq.	Coefficient of:							Right Side
		Z	x_1	x_2	x_3	x_4	x_5	x_6	
Z	(0)	1	2	0	0	1	1	0	8
x_2	(1)	0	5	1	0	1	3	0	14
x_6	(2)	0	2	0	0	0	1	1	5
x_3	(3)	0	4	0	1	1	2	0	11

x_1 x_4 x_5 are 0

b.) Defining equations are equation:

$$x_1 = 0$$

$$2x_1 - 2x_2 + 3x_3 = 5$$

$$x_1 + x_2 - x_3 = 3$$